α-$BaCu_2Se_2$ Thin-Film XRD Analysis and Bayesian Optimization

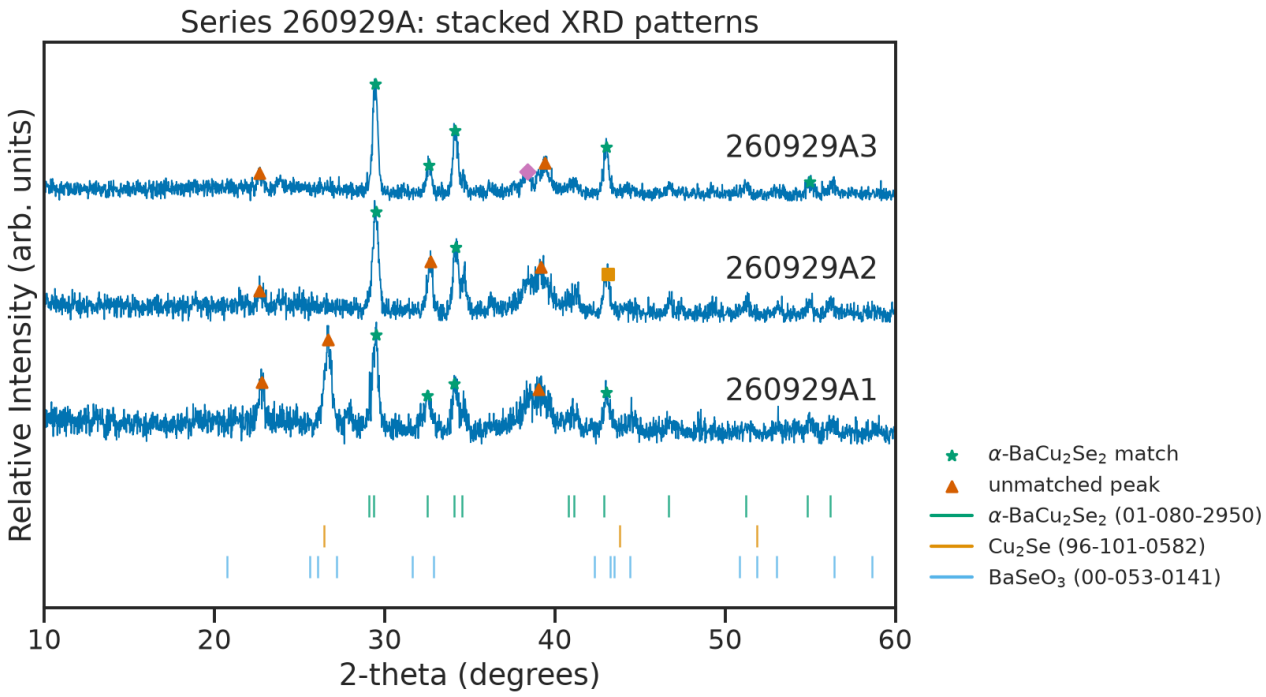
NOMAD-driven data extraction, phase matching, and GP-based synthesis planning

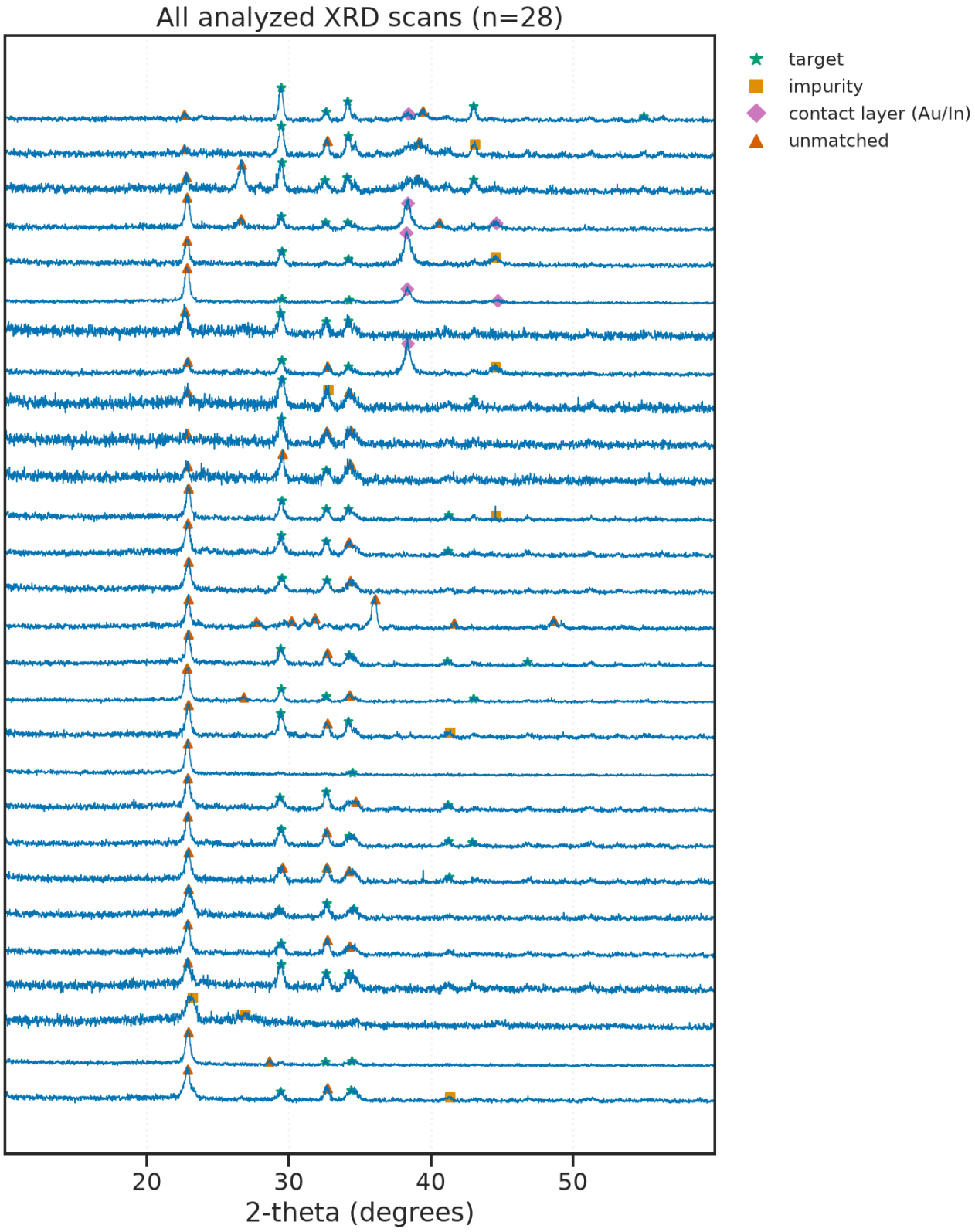
1. What was done

Synthesis records and XRD scans were pulled directly from the NOMAD Oasis REST API for the 42 α-$BaCu_2Se_2$ thin-film samples (INLThinFilmStack entries). Synthesis parameters (spin coating, soft/hard bake, selenization) and selenium-pulse settings were extracted for all samples; 35 include pulse on/off data. Each record also carries the precursor Cu/(Ba+Cu) ratio, the number of spin-coating cycles (deposited layers, excluding dynamic-dispense/seed-layer preparation steps), the Mo-coating flag, and the spin-to-selenization day gap. For the 28 films with XRD scans, diffraction patterns were matched against reference cards for the α-$BaCu_2Se_2$ target (01-080-2950), Cu_2Se (96-101-0582; α-Cu_2Se), and $BaSeO_3$ (00-053-0141). Au (96-901-1613) and In (96-151-2514) are classified separately as contact-layer matches with one shared plot symbol; their detected-peak intervals are excluded from phase-purity accounting and the unmatched/overlap penalty. In matching is restricted to series 260929A. Phase purity and crystallinity (1/FWHM of the strongest target match) were computed per film. A Gaussian-process Bayesian optimizer was trained on the 24 complete films to recommend the next synthesis conditions, optimizing objective = (0.7 × phase purity + 0.3 × normalized crystallinity) × (1 – 0.25 × A) over a five-input space: hard-bake temperature, selenization temperature and time, Cu/(Ba+Cu) ratio, and continuous Se pulse duty cycle. The on+off period is fixed at 3200 ms (3.2 s), so pulse frequency is fixed at 0.3125 Hz and is reported as metadata, not optimized. Intermediate duty cycles are model interpolations from three observed levels and should be treated as exploratory; recommended on-times are quantized to 1 ms increments. A long-form CSV of classified detected peaks is exported for sample-level audit. CBC is defined as $n_{Cu}/(n_{Ba}+n_{Cu})$; the candidate grid is 0.63–0.71, so a ratio of 1 (no Ba) is excluded. A is the unmatched/overlap share of detected peak-interval area, a screening proxy—not a quantitative phase fraction or a confirmed assignment of the ~23° feature.

2. XRD phase identification

Stacked XRD patterns per series with reference bars for the target and selected impurity phases. Markers flag target (star), impurity (square), shared Au/In contact (diamond), and unmatched (triangle) peaks. Au/In reference-card bars are omitted.

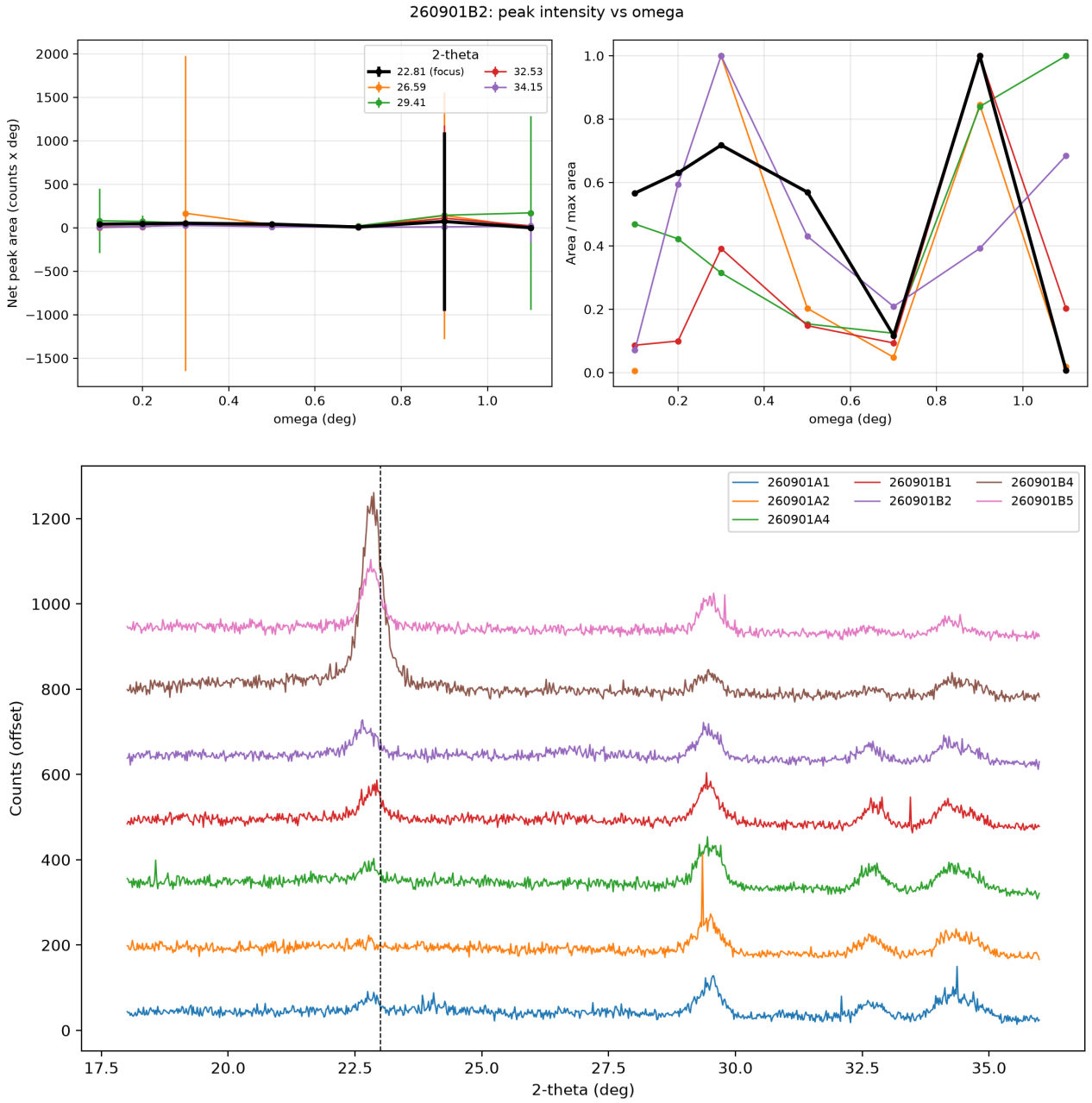

Series 260929A: stacked XRD patterns
Relative Intensity (arb. units)
260929A3
260929A2
260929A1
10
20
30
40
50
60
2-theta (degrees)
α-BaCu2Se2 match
unmatched peak
α-BaCu2Se2 (01-080-2950)
Cu2Se (96-101-0582)
BaSeO3 (00-053-0141)



Omega study of sample 260901B2

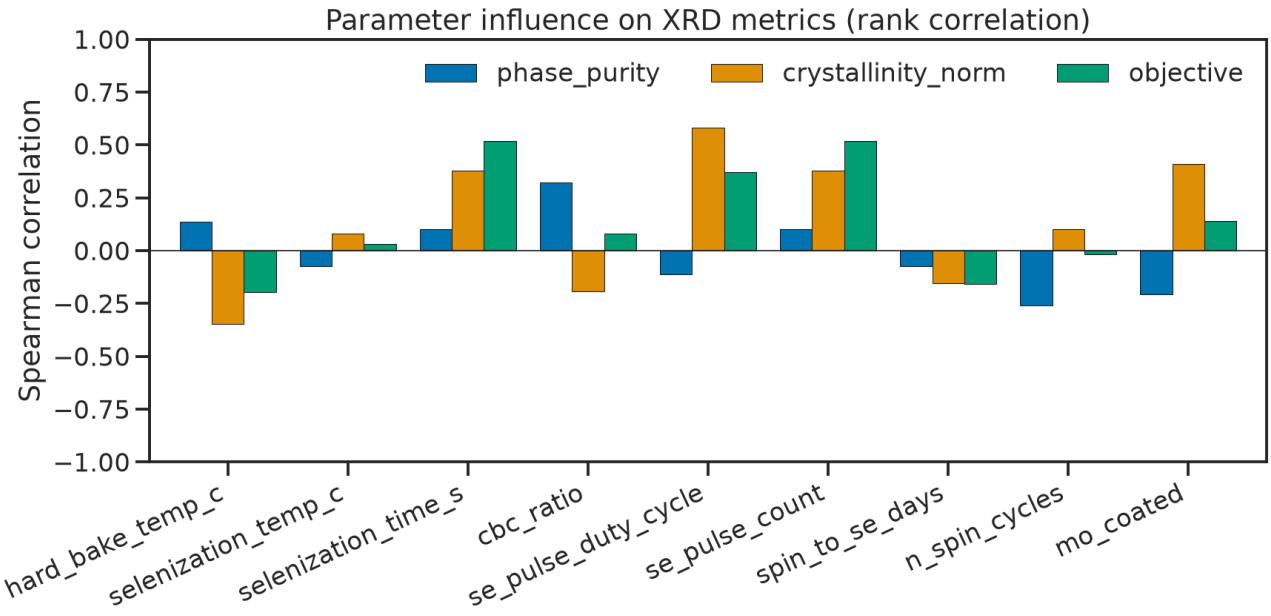
Seven grazing-incidence scans (omega 0.1–1.1°, 20–35° 2θ) were fitted peak by peak to test the origin of the ~23° feature (notebook 2.1; this sample only, not used by the optimizer). The feature sits at 22.81° 2θ with FWHM ≈ 0.4–0.5°, matches no reference card within the 0.15° tolerance (nearest: BaSe at 23.33°), and appears at 22.8–22.9° in all other 260901 films, including the new A1, A2 and A4, with strongly varying height. Since the SLG substrate shows no sharp peak at 23°, the feature originates in the film side: either an unlisted secondary phase or a $BaCu_2Se_2$-related reflection. Peak counts are low (~100) and fits

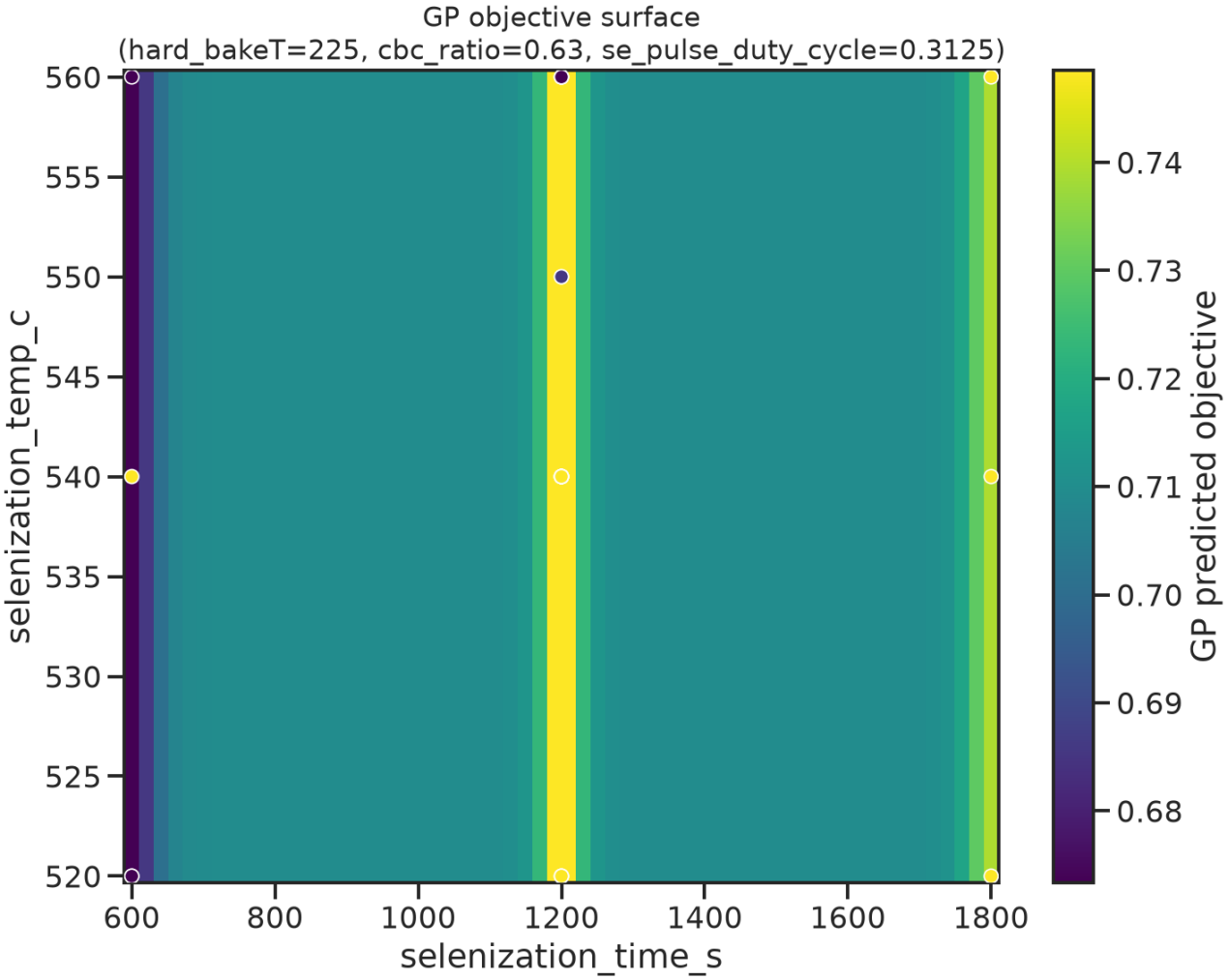
at omega ≥ 0.9° are unreliable, so the assignment remains unconfirmed; longer counting times and a candidate-phase search near 22.8° would help.



3. Main results

Across the analyzed films the mean phase purity is 0.93 and mean crystallinity is 2.16. The best film under the penalized score is 260929A3 (score 0.92; unpenalized score 0.99; purity 1.00, crystallinity 2.83, A 0.27). The optimizer uses the five selected inputs only. Other process and substrate descriptors remain available for descriptive exploration; duty cycle is modeled continuously, with the fixed pulse frequency excluded from the optimizer because it has no observed variation.



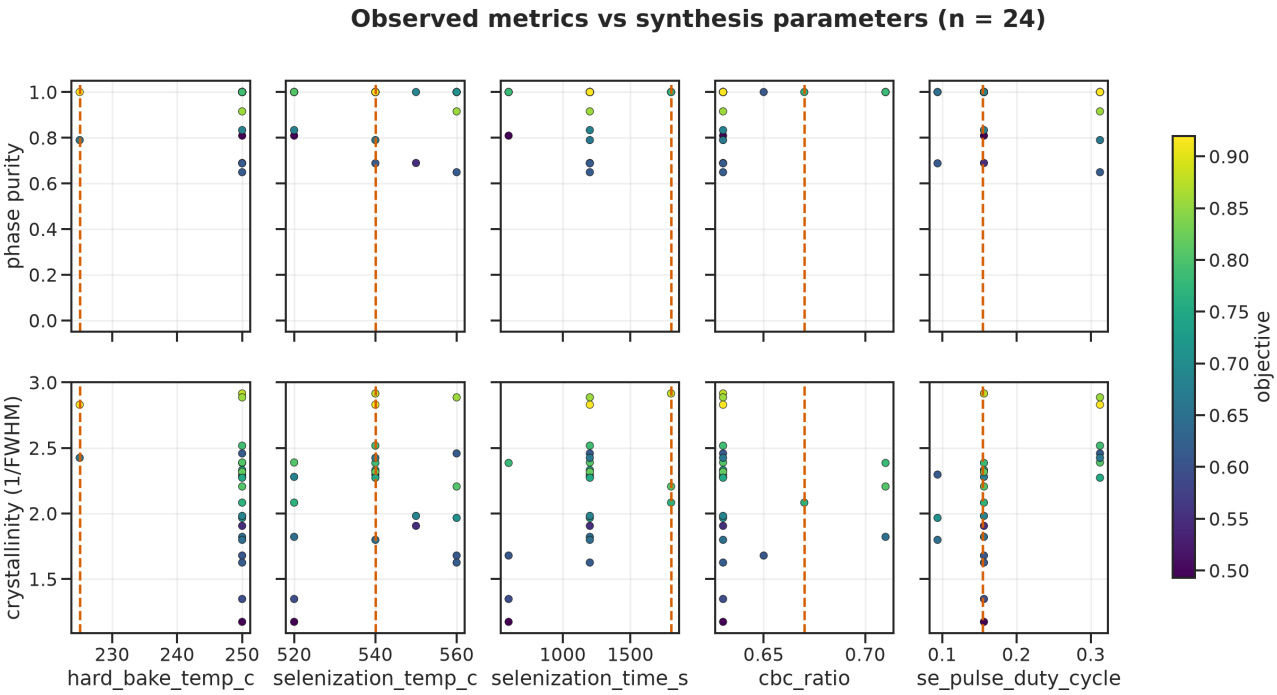


4. Suggested next experiments

Top Gaussian-process suggestions (predicted objective ± std), ranked by expected improvement:

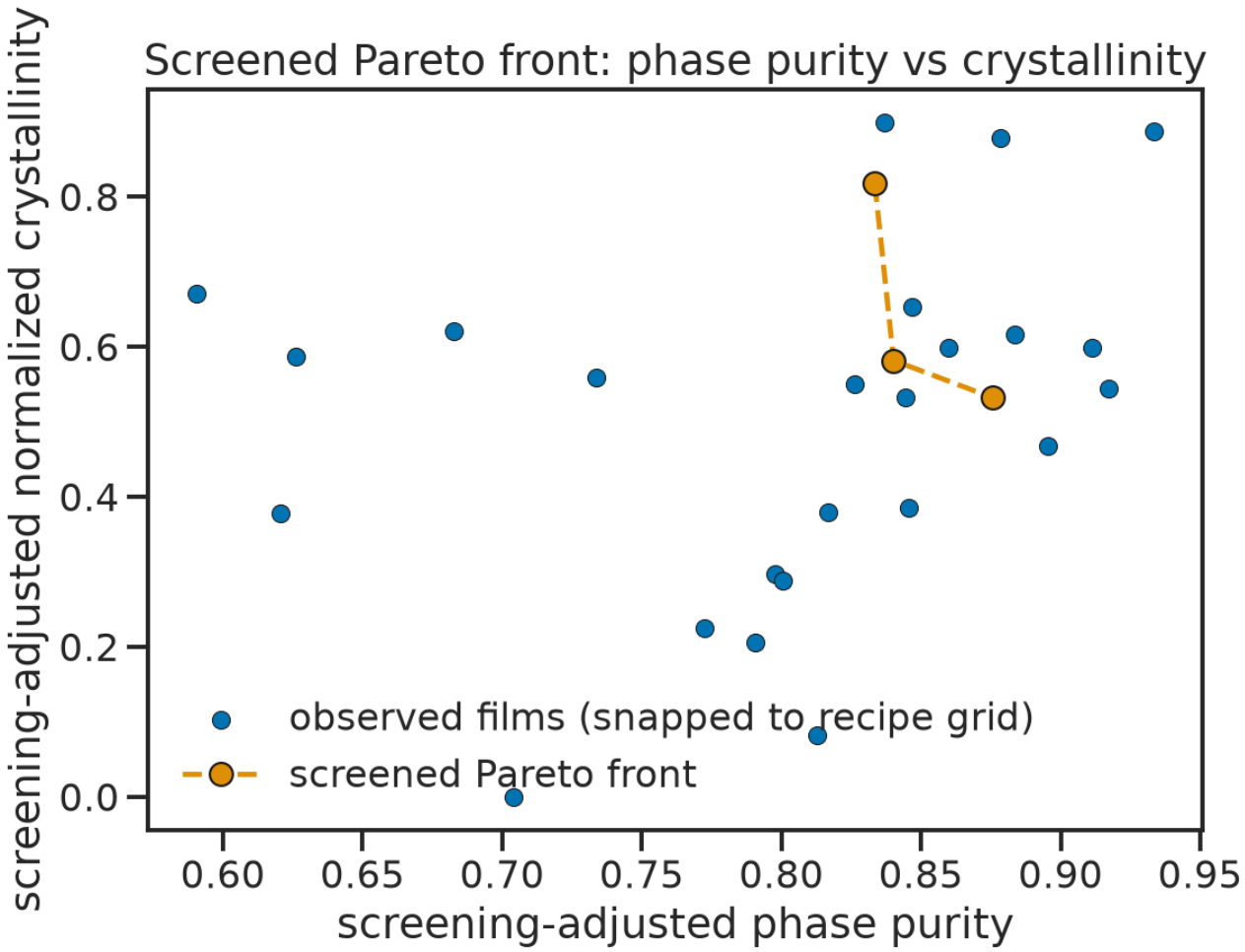
#	Hard bake °C	Selen. temp °C	Selen. time s	CBC ratio	Se duty cycle	Se on ms	Se off ms	Predicted score	GP std
1	225	540	1800	0.670	0.1547	495	2705	0.760	0.061
2	225	540	1800	0.670	0.1459	467	2733	0.710	0.068
3	225	540	1800	0.670	0.1353	433	2767	0.710	0.068
4	225	540	1800	0.650	0.1297	415	2785	0.710	0.068
5	225	540	1800	0.710	0.1278	409	2791	0.710	0.068

Recommended first experiment: hard bake 225 °C, selenization 540 °C for 1800 s, CBC ratio 0.670, and Se pulse duty cycle 0.1547 (495 ms on / 2705 ms off) (predicted objective 0.76).



5. Multi-objective optimization (Ax)

A complementary multi-objective search treats screening-adjusted phase purity and normalized crystallinity as two separate objectives: both are multiplied by the same (1 – 0.25 × A) factor. Ax (BoTorch) fits a Gaussian process per metric and finds the Pareto front via qNEHVI hypervolume maximization. Its five candidate inputs match notebook 3: hard-bake temperature, selenization temperature/time, CBC ratio, and continuous pulse duty cycle. Pulse period is fixed at 3.2 s (0.3125 Hz), so frequency is excluded as a constant. Duty-cycle suggestions are quantized to 1 ms on-time increments; interpolation between the three observed duty levels is exploratory. Numeric process recipes use the same explicit experiment-scale grids, including 25 °C bake steps. This remains a Pareto formulation, not the same scalar objective as notebook 3. A is an exploratory detected-peak-area screening proxy, not a phase fraction or a confirmed assignment of the ~23° feature.



#	Hard bake °C	Selen. temp °C	Selen. time s	CBC ratio	Se duty cycle	Se on ms	Se off ms
1	225	520	1800	0.630	0.1787	572	2628
2	250	520	1800	0.710	0.3125	1000	2200
3	225	560	1800	0.630	0.1316	421	2779
4	250	520	1800	0.630	0.2128	681	2519
5	225	560	1800	0.710	0.0938	300	2900